

Chapter-3 Line Integrals and Surface Integrals

This chapter covers -

- Line Integral of Scalar and Vector valued functions
- Concept of scalar potential and conservative forces
- Area of Curved Surfaces
- Surface Integrals of Scalar and Vector valued functions

Line Integral of a Scalar valued function

For a *scalar-valued* function $f(x, y, z)$, we denote its line integral over curve C by $\int_C f ds$.

To evaluate it -

First, parametrize the curve, i.e., Write the curve C as

$\vec{r}(t) = (x(t), y(t), z(t))$ for $t \in [a, b]$

Then, using this equation,

$$\int_C f(x, y, z) ds = \int_a^b f(\vec{r}(t)) \|\vec{r}'(t)\| dt$$

the line integral is converted into a friendly one-variable integral in t !

Important - We must have $a < b$

(Point to note - The value of the line integral does not depend on our parametrization of C)

Line Integral of a Vector valued function

Now, we have a *vector-valued* function $\vec{F}(x, y, z) = (P, Q, R)$. Here, P , Q and R are the components of $\vec{F}$, and are scalar valued functions themselves.

We denote the line integral of $\vec{F}$ over the curve C by $\int_C \vec{F} \cdot d\vec{s}$.

Steps to evaluate it -

Again, parametrize the curve! Write the curve C as

$$\vec{r}(t) = (x(t), y(t), z(t)) \text{ for } t \in [a, b]$$

Then, we can write the differential element as

$$d\vec{s} = \vec{r}'(t)dt = (x'(t), y'(t), z'(t))dt = (dx, dy, dz)$$

Writing the dot product, we get -

$$\int_C \vec{F} \cdot d\vec{s} = \int_C Pdx + Qdy + Rdz = \int_a^b (P, Q, R) \cdot (x'(t), y'(t), z'(t))dt$$

(Note $(Pdx + Qdy + Rdz)$ is called a *differential one form*)

Exercise 1

Find the line integral of

$$f(x, y, z) = x + y + z$$

over the straight-line segment from $(1, 2, 3)$ to $(0, -1, 1)$.

Exercise 1 - Solution

Choose the parametrization:

$$x(t) = 1 + (0 - 1)t = 1 - t$$

$$y(t) = 2 + (-1 - 2)t = 2 - 3t$$

$$z(t) = 3 + (1 - 3)t = 3 - 2t$$

where $t \in [0, 1]$.

$$\vec{r}(t) = (1 - t, 2 - 3t, 3 - 2t)$$

$$\vec{r}'(t) = (-1, -3, -2)$$

Exercise 1 - Solution

$$\begin{aligned}\int_C f(x, y, z) \, ds &= \int_0^1 (x(t) + y(t) + z(t)) \|\vec{r}'(t)\| \, dt \\&= \int_0^1 [(1 - t) + (2 - 3t) + (3 - 2t)] \|(-1, -3, -2)\| \, dt \\&= \sqrt{14} \int_0^1 (6 - 6t) \, dt = 3\sqrt{14}\end{aligned}$$

Exercise 2

Find the line integral of

$$f(x, y) = x^2 - y$$

over the curve

$$C : x^2 + y^2 = 4$$

in the first quadrant from $(0, 2)$ to $(\sqrt{2}, \sqrt{2})$

Exercise 2 - Solution

Choose parametrization,

$$\vec{r}(\theta) = (2 \cos \theta, 2 \sin \theta)$$

where θ goes from $\frac{\pi}{4}$ to $\frac{\pi}{2}$. (Note that the lower limit should necessarily be less than the upper limit)

$$\begin{aligned}\int_C f \, ds &= \int_{\pi/4}^{\pi/2} (4 \cos^2 \theta - 2 \sin \theta) \|(-2 \sin \theta, 2 \cos \theta)\| \, d\theta \\ &= 2 \int_{\pi/4}^{\pi/2} (2 + 2 \cos 2\theta - 2 \sin \theta) \, d\theta \\ &= \pi - 2 - 2\sqrt{2}\end{aligned}$$

Exercise 3

Find the work done by the force field

$$\mathbf{F} = (y - x^2)\mathbf{i} + (z - y^2)\mathbf{j} + (x - z^2)\mathbf{k}$$

along the curve

$$\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}, \quad 0 \leq t \leq 1,$$

from $(0, 0, 0)$ to $(1, 1, 1)$?

Exercise 3 - Solution

$$\vec{r}(t) = (t, t^2, t^3), \quad t \in [0, 1]$$

$$\vec{r}'(t) = (1, 2t, 3t^2)$$

$$\int_C \vec{F} \cdot d\vec{s} = \int_0^1 ((t^2 - t^2), t^3 - t^4, t - t^6) \cdot (1, 2t, 3t^2) dt$$

$$= \int_0^1 (2t^4 - 2t^5 + 3t^3 - 3t^8) dt$$

$$= \frac{2}{5} - \frac{2}{6} + \frac{3}{4} - \frac{3}{9} = \frac{29}{60}$$

Curl, Grad and all that!

We define the gradient of a scalar-valued function $f(x, y, z)$ as

$$\text{grad} f = \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

We define the curl of a vector-valued function $\vec{F} = (P, Q, R)$ as

$$\text{curl} F = \nabla \times \vec{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right)$$

A useful way to remember this is

$$\nabla \times \vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

We have the following important result -

$$\text{curl grad} f = \nabla \times \nabla f = 0$$

for all scalar valued functions f .

Scalar Potential of a Function

We say that a vector valued function $\vec{F}$ has a scalar potential ϕ , if

$$\vec{F} = \nabla\phi$$

If $\vec{F}$ has a scalar potential, it is a *Conservative* force. It satisfies the following properties -

$\vec{F}$ is conservative *iff* the work done in any closed loop is zero, i.e.,

$$\int_C \vec{F} \cdot d\vec{s} = 0$$

whenever C is a closed loop.

$\vec{F}$ is conservative *iff* work done is independent of path, and only depends on endpoints

We say that when $\vec{F}$ is conservative, the cooresponding differential one form ($Pdx + Qdy + Rdz$) is *exact*.

Condition for a force to be conservative

Consider $\vec{F} : G \rightarrow \mathbf{R}^3$, a vector valued function defined on open set G .
Then

$$\text{curl} \vec{F} = 0$$

is always a *necessary* condition for $\vec{F}$ to be conservative

Moreover, if G is acyclic of degree 1, then it is also a *sufficient* condition for $\vec{F}$ to be conservative, i.e., for existence of scalar potential.

Exercise 4

Find

$$\int_C (yz + e^x, xz + \cos y, xy + z) \cdot d\vec{s}$$

where C is the part of the unit circle in the x - y plane traversed from $(1, 0)$ to $(0, 1)$

Exercise 4

Find

$$\int_C (yz + e^x, xz + \cos y, xy + z) \cdot d\vec{s}$$

where C is the part of the unit circle in the x - y plane traversed from $(1, 0)$ to $(0, 1)$

(Hint - Check whether the integrand is conservative!)

Exercise 4 - Solution

We notice that the integrand is a conservative function with this scalar potential -

$$\phi = xyz + e^x + \sin y + \frac{z^2}{2} + C$$

Hence, using the property of conservative forces, we can directly write the line integral as

$$\begin{aligned} \int_C (yz + e^x, xz + \cos y, xy + z) \cdot d\vec{s} &= \phi(0, 1, 0) - \phi(1, 0, 0) \\ &= 1 - e + \sin 1 \end{aligned}$$

Note, that the path between $(1, 0, 0)$ and $(0, 1, 0)$ did not matter at all
 $\implies$ Path independence of work done by conservative forces!

Exercise 5

Evaluate the line integral

$$\oint_C \frac{x^2 y \, dx - x^3 \, dy}{(x^2 + y^2)^2}$$

where C is the square with vertices $(\pm 1, \pm 1)$ oriented in the anticlockwise direction.

Exercise 5 - Solution

Let us evaluate the line integral over each edge separately

$$I_1 = \int_{(1,-1)}^{(1,1)} \frac{x^2 y \, dx - x^3 \, dy}{(x^2 + y^2)^2}$$

Over this curve, let $\vec{r}(t) = (1, t)$ where t goes from -1 to 1 .

$$I_1 = \int_{-1}^1 \frac{(1, t) - (0, 1)}{(1 + t^2)^2} dt = \int_{-1}^1 \frac{-t \, dt}{(1 + t^2)^2} = -\frac{\pi}{4} - \frac{1}{2}$$

Exercise 5 - Solution

Now consider

$$I_2 = \int_{(1,1)}^{(-1,1)} \frac{x^2 y \, dx - x^3 \, dy}{(x^2 + y^2)^2}$$

On this segment, let $\vec{r}(t) = (t, 1)$ where t goes from 1 to -1

$$\therefore I_2 = \int_1^{-1} \frac{(t^2, -t^3) \cdot (1, 0)}{(t^2 + 1)^2} dt = \int_1^{-1} \frac{t^2 \, dt}{(t^2 + 1)^2} = \frac{-\pi}{4} + \frac{1}{2}$$

Similarly proceed for I_3 and I_4 .

$$\oint_C \frac{x^2 y \, dx - x^3 \, dy}{(x^2 + y^2)^2} = I_1 + I_2 + I_3 + I_4 = -\pi$$

Area of a Surface

To find the area of a given surface S , we first find a parametrization, $\vec{r}(u, v)$, such that $\vec{r}: R \rightarrow S$, where $R \in \mathbf{R}^2$ is the domain of $\vec{r}(u, v)$ that maps to the surface S .

Then the area of the surface is given by -

$$\text{Area} = \iint_R \|\vec{r}_u \times \vec{r}_v\| du dv$$

Surface integral of a Scalar valued function

Consider a scalar valued function $f(x, y, z)$ defined on the surface S . Then we can find its surface integral over that surface, denoted by $\iint_S f d\sigma$.

Steps to evaluate the surface integral -

Parametrize the surface! Determine the domain R of the parametrization. Then the surface integral is given by

$$\iint_S f(x, y, z) d\sigma = \iint_R f(\vec{r}(u, v)) \|\vec{r}_u \times \vec{r}_v\| du dv$$

This converts the surface integral into a simple double integral, which we know how to evaluate.

Surface integral of a Vector valued function

Consider a vector valued function $\vec{F}(x, y, z) = (P, Q, R)$ defined on the surface S . Then we can find its surface integral over that surface, denoted by $\iint_S \vec{F} \cdot d\vec{\sigma}$, also written as $\iint_S \vec{F} \cdot \vec{n} d\sigma$.

Steps to evaluate the surface integral -

Parametrize the surface! Determine the domain R of the parametrization. Then the surface integral is given by

$$\iint_S \vec{F}(x, y, z) \cdot d\vec{\sigma} = \iint_R \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) du dv$$

The surface integral of a vector valued function is effectively the "flux" of the vector function over the surface.

Area of surface of revolution and Pappus Theorem

Consider a curve lying $\vec{r}(t) = (x(t), y(t))$ lying in the x-y plane. Now revolve this curve about the x-axis, which gives us the surface of revolution of this curve. What is the surface area of this curve?

First of all, the parametrization of that surface is given by -

$$\vec{r}(t, \theta) = (x(t), y(t)\cos\theta, y(t)\sin\theta)$$

Theorem (Pappus theorem)

$$\text{Area} = 2\pi \int_C y ds = 2\pi IY$$

where I is the arc length of the curve and

Y is the y-coordinate of the "centre of mass" of the curve

Exercise 6

Compute the area of that portion of the paraboloid

$$x^2 + z^2 = 2ay$$

which is between the planes $y = 0$ and $y = a$.

Exercise 6 - Solution

Choose parametrization

$$\vec{r}(\rho, \theta) = (\rho \cos \theta, \frac{\rho^2}{2a}, \rho \sin \theta)$$

$$\vec{r}_\rho = \left(\cos \theta, \frac{\rho}{a}, \sin \theta \right) \quad \text{and} \quad \vec{r}_\theta = (-\rho \sin \theta, 0, \rho \cos \theta)$$

$$\vec{r}_\rho \times \vec{r}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \theta & \frac{\rho}{a} & \sin \theta \\ -\rho \sin \theta & 0 & \rho \cos \theta \end{vmatrix} = \left(-\frac{\rho^2 \cos \theta}{a}, -\rho, \frac{\rho^2 \sin \theta}{a} \right)$$

$$\|\vec{r}_\rho \times \vec{r}_\theta\| = \rho \sqrt{1 + \frac{\rho^2}{a^2}}$$

$$\therefore \text{Area} = \int_0^{2\pi} \int_0^{a\sqrt{2}} \rho \sqrt{1 + \frac{\rho^2}{a^2}} d\rho d\theta$$

Exercise 7

Compute the area of the surface of revolution:

$$\vec{r}(u, v) = u \cos v \hat{i} + u \sin v \hat{j} + u^2 \hat{k}$$

$$0 \leq u \leq 4, \quad 0 \leq v \leq 2\pi$$

Exercise 8

Find the center of mass of that portion of the homogeneous hemispherical surface

$$x^2 + y^2 + z^2 = a^2$$

lying above the first quadrant in the xy -plane.

Exercise 9

Find the flux of the field

$$\mathbf{F}(x, y, z) = z^2\mathbf{i} + x\mathbf{j} - 3z\mathbf{k}$$

upward through the surface cut from the parabolic cylinder

$$z = 4 - y^2$$

by the planes $x = 0$, $x = 1$, and $z = 0$.

Odds and Ends : Integrating Factors

We have discussed what it means when a differential one form is *exact*
In cases where it is not exact, we may be able to *make it* exact by multiplying with a scalar-valued function $\lambda(x, y, z)$

So, for a differential one form $Pdx + Qdy + Rdz$,
 $\lambda(x, y, z)$ is called its **integrating factor**, if $\lambda(x, y, z)(Pdx + Qdy + Rdz)$ is exact, and $\lambda(x, y, z) > 0 \forall (x, y, z)$.
(Important - note the condition that $\lambda(x, y, z) > 0 \forall (x, y, z)$)

Odds and Ends : Integrating Factors

Recall that this is exactly what we did to solve linear differential equations in JEE! We multiplied both sides of the equation with a term called the Integrating factor, which made the LHS "exact". That allowed us to write the LHS as -

$$\lambda \frac{dy}{dx} + \lambda Py = \frac{d}{dx}(\lambda y)$$

which is to say

$$\lambda dy + \lambda Py dx = d(\lambda y)$$

where λ is the integrating factor

Odds and Ends : Integrating Factors

Coming back to the matter at hand, if we have $\vec{F} = (P, Q, R)$

Since multiplying by the integrating factor makes the differential one form of $\vec{F}$ exact, (which is to say $\lambda(Pdx + Qdy + Rdz)$ is exact),

We can conclude that $\lambda\vec{F}$ must be *conservative*.

This implies $\text{curl}(\lambda\vec{F}) = 0$

This can be simplified to (refer to proof in the notes)

$$\vec{F} \cdot \text{curl}\vec{F} = 0$$

Exercise 10

Show that $(xyz)^{-2}$ is an integrating factor for $yz(y+z) dx + zx(z+x) dy + xy(x+y) dz$ in the positive octant $(x > 0, y > 0, z > 0)$.

Exercise 10 - Solution

$$Pdx + Qdy + Rdz = yz(y + z) dx + zx(z + x) dy + xy(x + y) dz$$

$$\text{Let } \lambda = (xyz)^{-2}$$

$$\therefore \lambda(Pdx + Qdy + Rdz) = \frac{(y + z) dx}{x^2 yz} + \frac{(z + x) dy}{xy^2 z} + \frac{(x + y) dz}{xyz^2}$$

$$\text{Consider the function } \vec{F}_1 = \left(\frac{y + z}{x^2 yz}, \frac{z + x}{xy^2 z}, \frac{x + y}{xyz^2} \right)$$

Exercise 10 - Solution

Find its scalar potential

$$\phi = -\frac{1}{xz} - \frac{1}{xy} + \phi_1(y, z)$$

$$\therefore \phi = -\frac{1}{xy} - \frac{1}{yz} - \frac{1}{zx} + C \quad \text{is the scalar potential}$$

Since the scalar potential exists, we can write

$$\lambda(Pdx + Qdy + Rdz) = d\phi = d\left(-\frac{1}{xy} - \frac{1}{yz} - \frac{1}{zx} + C\right)$$

Hence $\lambda(Pdx + Qdy + Rdz)$ is exact. So, λ is an integrating factor.

Hence proved

All the best!